

LLM Output Variance Analysis Dashboard

Documentation

1. Overview

The **LLM Output Variance Analysis Dashboard** is an internal evaluation tool for the Schaeffler Mobility Insight Platform. It is designed to help researchers and developers understand how consistent (or variable) a large language model (LLM) is when answering the **same query** multiple times. By logging every run of the application and analyzing the variability of the outputs, the dashboard provides quantitative insight into model stability, identifies prompts that may need refinement, and tracks whether improvements (e.g., new prompts or model versions) lead to more reliable results.

Why This Matters

Unlike traditional software functions, LLMs are inherently non-deterministic: when you feed the exact same input, you can receive different outputs. Variability is sometimes desirable (it allows for creativity), but in a professional context—such as generating trend forecasts, assessments, and implementation plans—too much variability erodes trust. The evaluation dashboard allows Schaeffler’s innovation team to:

- **Measure stability** across multiple trials for the same query.
- **Compare variability** between different output sections (e.g. trend generation, assessment, market plan).
- **Identify problematic prompts** or sectors where the LLM produces inconsistent recommendations.
- **Track improvements over time** after changing prompts, models or parameters.

2. Data Sources and Logging

The dashboard reads directly from the `trend_queries` table in the Mobility Insight Platform’s database. Each row represents a complete run of the application for a specific combination of:

- **Use Case** – the high-level problem or user story (e.g. *People mover mobility*).
- **Sector** – the market segment or industry vertical (e.g. *RoboTaxi*).
- **Demand** – the primary user request or desired outcome (e.g. *navigation optimisation*).

For each run, the table also stores the output for each stage: trend generation, assessment, radar positioning, PESTEL relation, implementation plan, partners, as well as the **selected trend**, the **confidence score** (if available) and the **timestamp**. These logs are appended automatically whenever a user interacts with the application, so the dashboard can analyze historical data without any modification to the core app.

3. Methodology

3.1 Query Fingerprinting

To analyze output variability for the same input, the tool groups historical runs by **query fingerprint**. A fingerprint is a short hash (the first twelve characters of an MD5 digest) of the string `use_case|sector|demand`. All runs with the same fingerprint correspond to identical user questions and are considered **trials** of the same query.

3.2 Parsing and Metric Extraction

Each run contains free-form text generated by the LLM. The dashboard uses regular expressions to extract important elements:

- **Trend titles** – identified by patterns like `Technology Title: ###` or `Trend n: ###`.
- **Confidence scores** – numerical values following `Confidence Score: ###`.
- **Classification** – whether radar positioning labelled the trend as `ACT`, `PREPARE` or `WATCH`.
- **Budgets & ROI** – currency figures (e.g. “€ 1.5 million”) and percentages (e.g. “12%”) from the implementation plan.
- **Assessment scores** – numbers in the assessment matrix (e.g. “Synergy: 8/10”).

These extracted values allow the dashboard to compute structured metrics on otherwise unstructured text without requiring model re-training. When a certain field is missing or extraction fails, the metric is simply left blank for that trial.

3.3 Variability and Stability Metrics

For each group of trials with the same fingerprint, the following metrics are computed:

Metric	Description	Interpretation
Trend Diversity	Ratio of unique trend titles to total trend titles across trials.	High diversity (>80%) means the LLM is generating many new ideas; low diversity (<20%) means the outputs are very similar.
Confidence CV	Coefficient of variation of confidence scores. CV = standard deviation ÷ mean.	Lower CV (<15%) implies stable scoring; higher CV (>30%) suggests inconsistent confidence valuations.
Text Similarity	Mean pair-wise Jaccard similarity of the trend description texts (word overlap).	A high similarity (>70%) indicates the narrative content remains largely the same; low similarity (<40%) means the descriptions vary significantly.

Metric	Description	Interpretation
Budget & ROI Variance	Average and standard deviation of extracted budget figures and ROI percentages.	Large variance warns that the cost projection or ROI fluctuates drastically across runs.
Classification Entropy	Entropy of the radar classification across trials, computed as $-\sum p_i \log_2(p_i)$ where p_i is the fraction of trials labelled <code>ACT</code> , <code>PREPARE</code> or <code>WATCH</code> .	Entropy near zero means consistent classification; higher entropy means the LLM changes its decision between <code>ACT</code> , <code>PREPARE</code> and <code>WATCH</code> .
Temporal Trend	Regression slope of weekly variance (e.g., confidence std) over time.	Detects whether outputs are stabilising or becoming more volatile as prompts/models change.

3.4 Dashboard Structure

The dashboard is a Streamlit application with six tabs:

1. **Variance Overview** – A bar-chart overview showing diversity, confidence CV, budget CV and text consistency for each query that meets the trial threshold. It highlights high-variance queries and points out highly consistent or diverse outputs.
2. **Query Deep Dive** – Allows selection of a single query fingerprint. It displays detailed statistics (e.g. diversity ratio, confidence mean & std), a trial-by-trial comparison table, a line chart of confidence evolution and a heatmap of text similarities. This helps to identify exactly how outputs differ across runs.
3. **Temporal Analysis** – Aggregates metrics on a weekly basis. It visualises trial volume, average confidence with error bars and unique trends over time, and performs a simple linear trend test to detect whether variance is increasing or decreasing.
4. **Consistency Metrics** – Produces box plots comparing consistency for trends, assessments and market plans across queries; shows correlations between number of trials and consistency; provides a scatter plot with bubble size representing classification entropy; and lists queries that are highly consistent, highly variable or exhibit mixed behaviour.
5. **Raw Data** – A table viewer that allows filtering by fingerprint and selecting columns. Users can download the filtered subset as CSV for offline analysis. Summary statistics of coverage and data quality are also displayed.
6. **Full History** – A history explorer that lists every stored query and its answers. Users can filter by date range, use case, sector and demand, group records by fingerprint, and download full or group-level datasets. This is useful for audits and reproducibility.

4. Using the Dashboard

1. **Load data:** The dashboard automatically loads all historical runs from `trend_queries`. Make sure the database credentials (host, user, password, database name) are set in environment variables or Streamlit secrets.
2. **Set minimum trials per query:** Use the sidebar slider to exclude queries with too few runs. Variability metrics are only meaningful when there are at least two trials.

3. **Filter by use case:** Select specific use cases to narrow the analysis. This reduces noise and focuses the charts on relevant domains.
4. **Interpret the overview:** Look for queries with high CV or low similarity; they may need prompt adjustments or additional guidance to the LLM.
5. **Drill down:** Use the deep dive tab to inspect problematic queries. Compare text outputs, examine how confidence scores evolve, and see if certain trends repeat.
6. **Check trends over time:** Use the temporal analysis to see whether variance metrics improve after deploying new prompts or models. A decreasing slope suggests progress.
7. **Validate consistency:** Use the consistency metrics tab to identify which output sections are stable and which are not. Large bubbles in the scatter plot indicate inconsistent radar classification.
8. **Access raw data:** The raw data and history tabs let you inspect and export the underlying records for further research or auditing.

5. Outcomes and Interpretation

The evaluation dashboard transforms raw logs into actionable insights. A few typical outcomes:

- **Stable queries:** Low diversity, low confidence CV and high similarity indicate that the LLM reliably returns the same trends and valuations across runs. These are safe for production use without manual review.
- **Volatile queries:** High confidence CV, low text similarity and high budget variance highlight queries for which the LLM behaves unpredictably. They require prompt refinement or additional constraints (e.g. always output the same number of trends, provide numeric ranges).
- **Trend vs. Implementation:** The dashboard often shows that trend generation may be stable while the implementation plan (budgets, ROI) is highly variable. This suggests focusing prompt engineering on the later stages.
- **Temporal improvements:** If new prompts reduce variance or increase consistency over weeks, the slopes in temporal analysis will be negative or trending towards stability. Otherwise, the team needs to investigate whether modifications inadvertently introduced more randomness.

6. Limitations and Extensions

While the current dashboard provides a robust evaluation of output variance, there are a few limitations and possible extensions:

- **Semantic similarity:** The Jaccard similarity used here considers word overlap but not meaning. Two texts may use different words to describe the same concept and thus appear dissimilar. Future versions could integrate sentence embeddings to capture semantic similarity.
- **Regex dependency:** Extracting metrics using regular expressions may fail if the LLM's formatting changes. Standardising output into JSON or using a structured prompt would make parsing more reliable.
- **Versioning:** The dashboard does not yet separate runs by model version or prompt version. Adding columns like `model_name`, `prompt_version` or `app_version` to the log table would allow more granular comparisons.
- **Interactivity:** The tool currently provides read-only visualisations. A future "feedback" layer could allow users to tag trials as good or bad and integrate that qualitative feedback into the metrics.

7. Conclusion

The LLM Output Variance Analysis Dashboard is a vital component of the Schaeffler Mobility Insight Platform. It enables data-driven evaluation of large language model outputs, ensuring that the company's innovation insights are both reliable and actionable. By systematically measuring diversity, stability and consistency across multiple runs and over time, the dashboard empowers prompt engineers and product owners to make informed decisions, refine their workflows, and maintain high standards in mobility trend forecasting and implementation planning.
